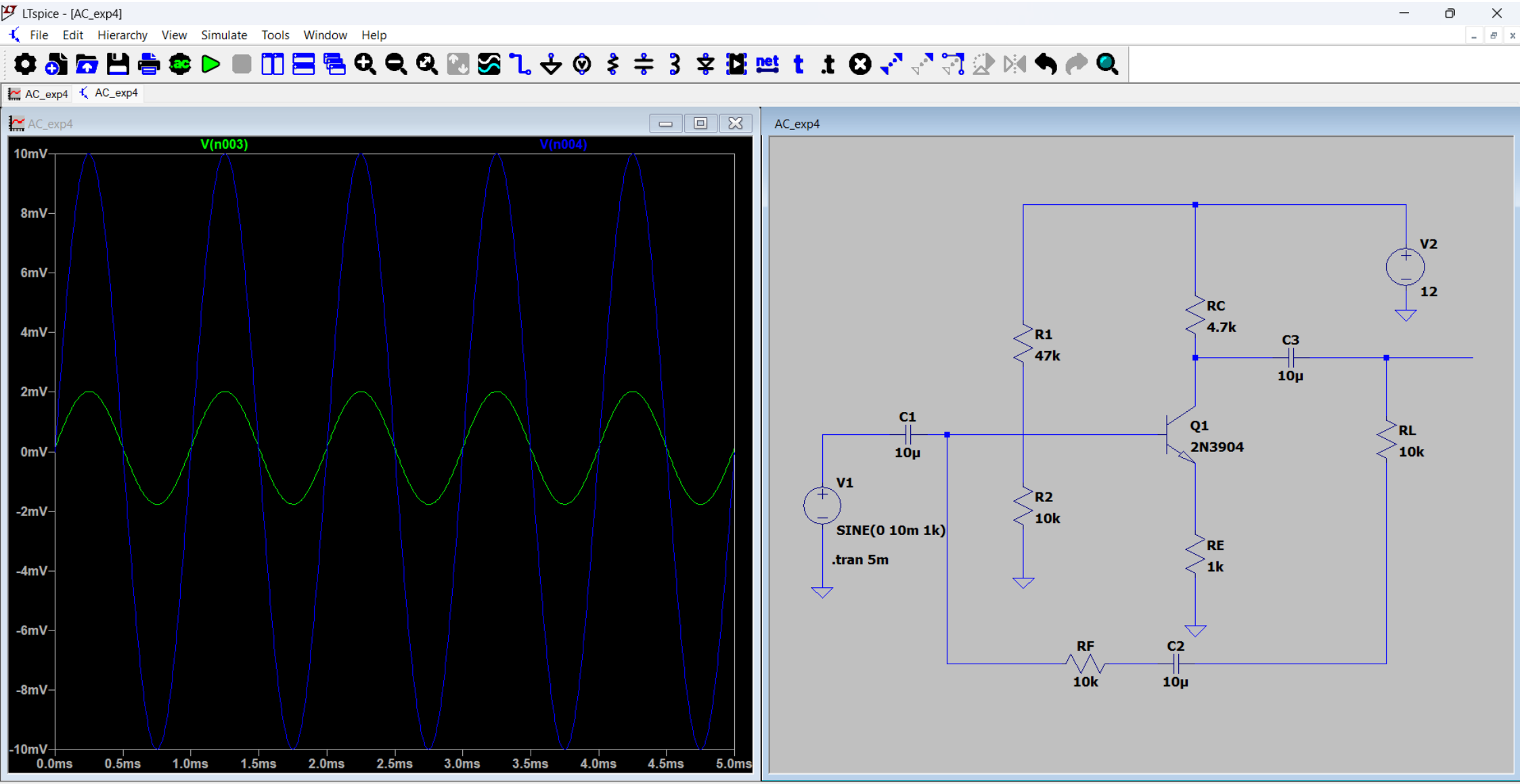
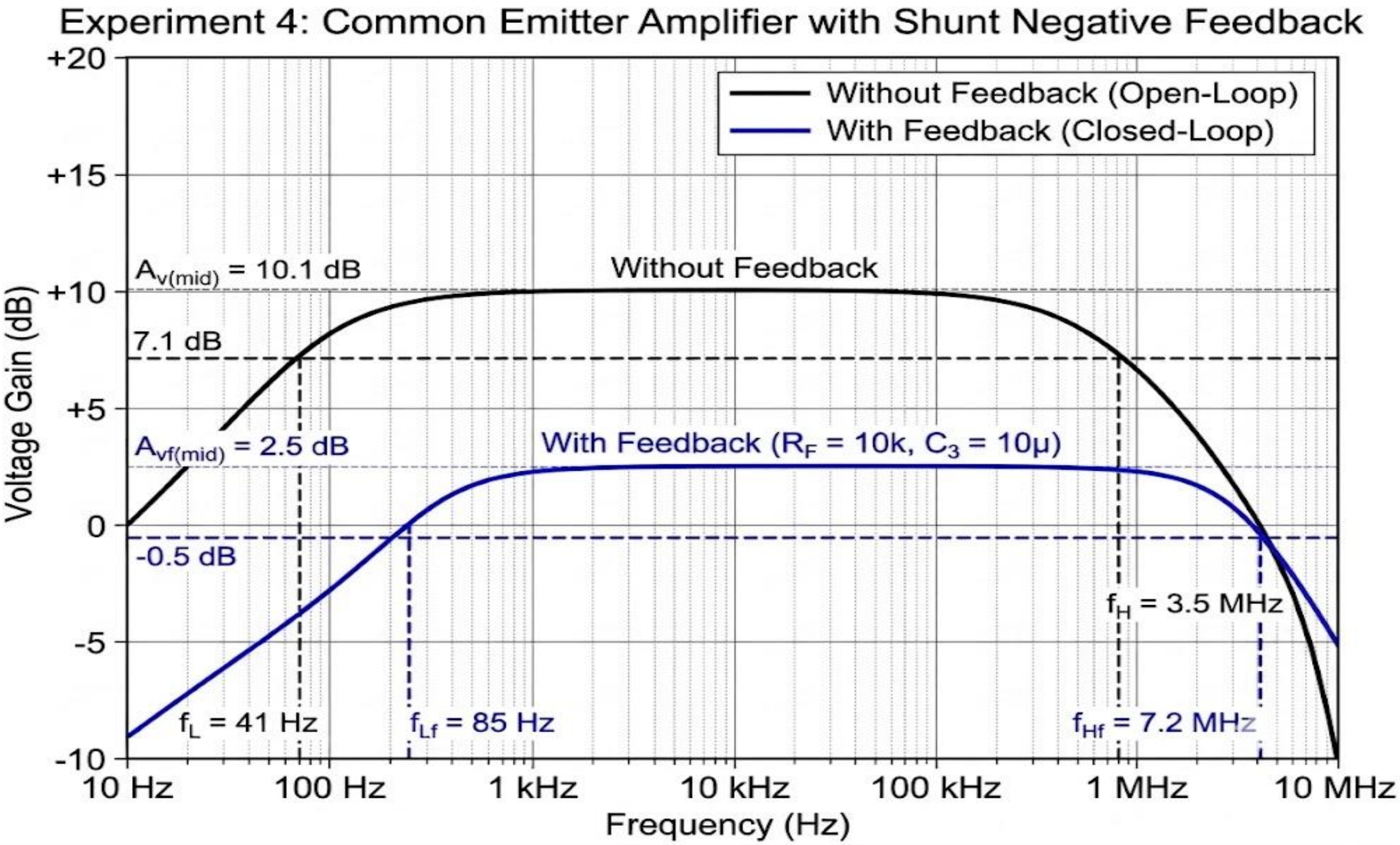

LTspice - [AC_exp4]
File Edit Hierarchy View Simulate Tools Window Help
AC_exp4
AC_exp4
V(n003)
V(n004)
10mV
8mV
6mV
4mV
2mV
0mV
-2mV
-4mV
-6mV
-8mV
-10mV
0.0ms 0.5ms 1.0ms 1.5ms 2.0ms 2.5ms 3.0ms 3.5ms 4.0ms 4.5ms 5.0ms
AC_exp4
V2
12
RC
4.7k
R1
47k
C3
10µ
C1
10µ
Q1
2N3904
RL
10k
V1
SINE(0 10m 1k)
.tran 5m
R2
10k
RE
1k
RF
10k
C2
10µ


Experiment 4: Common Emitter Amplifier with Shunt Negative Feedback
Without Feedback (Open-Loop)
With Feedback (Closed-Loop)
+20
+15
+10
+5
0
-5
-10
Voltage Gain (dB)
$A_{v(mid)}$ = 10.1 dB
Without Feedback
7.1 dB
$A_{vf(mid)}$ = 2.5 dB
With Feedback (R_F = 10k, C_3 = 10µ)
-0.5 dB
f_H = 3.5 MHz
f_L = 41 Hz
f_{Lf} = 85 Hz
f_{Hf} = 7.2 MHz
10 Hz
100 Hz
1 kHz
10 kHz
100 kHz
1 MHz
10 MHz
Frequency (Hz)